

Block E: capstone (author case)

Predicting crypto “zombies”

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MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



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Papers and materials

Będowska-Sójka, B., Wójcik, P., & Pele, D. T. (2025).

Early warning systems for cryptocurrency markets: Predicting 'zombie' assets using machine learning.

The North American Journal of Economics and Finance, 81, 102543.

DOI [10.1016/j.najef.2025.102543](https://doi.org/10.1016/j.najef.2025.102543)

Paper card: Będowska-Sójka, Wójcik & Pele (2025)

	<i>North American Journal of Economics and Finance 81: 102543</i>
Question	Which cryptocurrencies will become "zombies" (still listed but untradable) within the next 28 days?
Data	Weekly snapshots of cryptocurrencies listed for at least 210 days, 2015–2022; return, volume and market-cap features.
Method	Classical econometric baselines vs tree-based ML (random forests, boosting), with class-imbalance handling (downsampling, ROSE, SMOTE) and explainable-AI tools (permutation importance, partial dependence).
How it is tested	Out-of-time, walk-forward evaluation; balanced accuracy.
Main result	Random forests beat the classical models, with balanced accuracy of about 84% out of time. Volumes and past returns matter most.
Watch for	An early warning, not a price forecast: which error costs more, a missed zombie or a false alarm?
Code & data	Quantinar course and QuantLet code with the weekly snapshot CSV; some auxiliary files are not public.

Why this module closes the day

- Block D treated replication **from the outside**; here we see it **from the author's side**
- Live **author** paper: design choices you can defend
- Classification → **risk decision** (not alpha long–short)
- **Replicability**: Quantinar + QuantLet (public code; some aux files are not public)
- Adds **XAI** (importance, PDPs). Remember the opener: importance is not $\hat{\beta}$

Economic question

Which cryptocurrencies are about to become **zombies** — still “existing” but **untradable** long enough that investors cannot exit?

Early warning $\neq$ price forecast.

Label (paper)

- Horizon: zombie within the next **28 days**
- Sample spirit: assets listed long enough to learn history (paper: ≥ 210 days; 2015–2022)
- Discuss: definition of untradable / exchange coverage / delisting

Features

- Return statistics
- Trading volume
- Market capitalisation
- Asset-specific characteristics

Information set: known before the warning window.

Paper finding to preview: **volumes** and **past returns** matter most.

Validation & results (headline)

- **Out-of-time** evaluation
- **Balanced accuracy** $\approx 84\%$ OOT (paper)
- Tree-based models (esp. **random forests**) beat classical econometrics
- Imbalance handling appears in the QuantLet scripts (downsampling / ROSE / SMOTE)

Model ladder → XAI

classical baselines → RF / boosting (xgb) → explain best model

QuantLet highlights:

- Code/ — training variants + 04_summarize_the_results.R
- 5_XAI_for_best*.R — permutation importance / PDP-style analysis

Decision costs (discussion)

Error	Investor / desk consequence
False negative	capital stuck in untradable asset
False positive	unnecessary exit / missed upside

Same Lessmann lesson: pick the metric that matches the **loss**.

Live materials

Where	What
Quantinar · Cryptos Zombies	PDF + video (optional backdrop)
<code>notebooks/05_crypto_zombies_csv_only.ipynb</code>	Live classroom demo (CSV only)
<code>experiments/Crypto_Zombies/workdir/.../all_symbols.csv</code>	public QuantLet data
Full <code>Code/*.R</code>	skip live (needs aux files)

Map to the rest of the day

Earlier module	Bridge
Krauss	classification scores → action
Credit (C)	default / early-warning flavour
Lessmann	fixed protocol + baseline discipline
Gu / Chen	different object (returns / SDF)
Replication (D)	what the public code reproduces, and what it does not

Replication from the author's side

Question	This paper
What is public?	QuantLet: training variants, result summary, XAI scripts; weekly snapshot CSV
Does it run end to end?	No: the aux xlsx and <code>F_summary_*.R</code> are not in the public archive
Robustness in the package?	Not packaged separately
Where will classroom numbers differ?	single OOT window vs walk-forward

Exercise (5 min): design the robustness appendix this replication package should contain.

Do differently

1. 14 vs 28 vs 60-day zombie horizon
2. Explicit cost-sensitive threshold
3. Size / exchange filters
4. Require stable XAI stories before model changes

Takeaway

Crypto zombies are still the craft of the day:

label → information set → out-of-time test → decision costs → explain → replicate.

Next: Cakici (`07_cakici.md`), then wrap (`08_wrap.md`)